

MAKE UP EXAMINATIONS - MARCH 2023

Program	: B.E. - Information Science and Engineering	Semester	: V
Course Name	: Computer Networks	Max. Marks	: 100
Course Code	: IS51	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- List the various causes for fragmentation. Also distinguish between the two different types of fragmentation CO1 (06)
 - Suppose a router has built up the routing table shown below. The router can deliver packets directly over interfaces 0 and 1, or it can forward packets to routers R1, or R2. Describe what the router does with a packet addressed to each of the following destinations: (Calculation need to be shown for every address). CO1 (08)
 - 135.46.63.10
 - 135.46.57.14
 - 135.46.52.2
 - 192.53.40.7
 - 192.53.56.7

Routing Table

Address/mask	Next hop
135.46.56.0/22	interface 0
135.46.60.0/22	interface 1
192.53.40.0/23	router 1
default	router 2

- An organization is granted the block 202.16.170.0/24. The administrator wants to create 24 subnets. CO1 (06)
 - Find the subnet mask.
 - Find the number of address in each subnet.
 - Find the first and last address in each subnet 1
 - Find the first and last address in each subnet 24.
- Demonstrate Placement and operation of a NAT box in detail. CO1 (06)
 - Explain Dual Stack and Tunneling with neat diagrams? CO1 (08)
 - Discuss the Classful addressing in detail. CO1 (06)

UNIT - II

3. a) Use the Dijkstra's algorithm and find the shortest path from node A to all other nodes. Clearly give the shortest path trees for all the iterations and final shortest path table: CO2 (08)

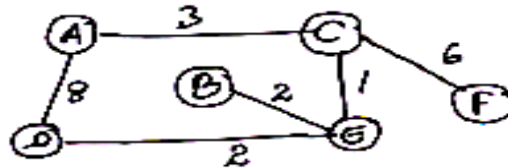


Fig.3(a)

- b) Describe two-node loop instability problem in distance vector routing protocol with an example. State the solutions for the same. CO2 (06)
- c) Illustrate different packet formatting techniques with an example. CO2 (06)
4. a) State the limitations of RARP. Discuss, how it is resolved in BOOTP. CO2 (07)
- b) Discuss the working of DHCP with a neat diagram. CO2 (06)
- c) Discuss the different decision making strategies in DVMRP. CO2 (07)

UNIT - III

5. a) Illustrate with necessary diagrams, the different mechanisms of retransmission supported in TCP to handle errors while transmission of segments. CO3 (06)
- b) Differentiate between connection oriented service and connectionless service. CO3 (04)
- c) Host A wants to send data to Host B. Host A produces bursty traffic as follows; CO3 (06)
- ✓ 8 data packets for first 2 seconds
 - ✓ 15 data packets for next 2 seconds
 - ✓ No data for next 3 seconds
 - ✓ Finally 13 data packets in the last second
- Leaky bucket technique with bucket size = 12 packets and out rate = 5 packets per second is implemented in between the two hosts A and B for shaping the traffic. Assuming all the packets to be of same size, enumerate the working of leaky bucket technique for the above scenario. Clearly mention how many packets are sent into the network, retained in the bucket and dropped in every time frame (in seconds).
- d) Describe the working of priority queuing with a neat diagram. What is the drawback of priority queuing? Which scheduling technique overcomes this drawback? CO3 (04)
6. a) Illustrate the three phases in handling of congestion by TCP with flowchart and an example. CO3 (10)
- b) Differentiate between a TCP segment and SCTP Packet. CO3 (05)
- c) Illustrate with diagram, why network congestion occurs in a system that involves routers with queues. CO3 (05)

UNIT- IV

7. a) What is Resolution? Discuss recursive and iterative resolution techniques to handle name-address resolution in DNS. CO4 (05)
- b) Describe the services provided by user agent to the user to make the process of sending and receiving a message easier. CO4 (10)
- c) What are the different FTP transmission modes? Differentiate between FTP and Anonymous FTP. CO4 (05)
8. a) Discuss FQDN and PQDN. Determine which of the following is FQDN & PQDN; i)xxx.yyy.net. ii)xxx.yyy. CO4 (05)
- b) Describe the different functions of network management system in brief. CO4 (10)
- c) Illustrate usage of different NVT character set for option negotiation with an example. CO4 (05)

UNIT - V

9. a) Illustrate symmetric and asymmetric key cryptography with a block diagram. CO5 (06)
- b) Enumerate the step-by-step procedure of RSA encryption algorithm. CO5 (08)
- c) Describe the various services provided by the network security. CO5 (06)
10. a) Explain the process of key exchange in Diffie-Hellman key exchange algorithm, illustrate the process with an example. if $g=7$, $p=23$, $x=3$ and $y=7$? CO5 (08)
- b) Illustrate the two way of achieving digital signature with a block diagram. CO5 (06)
- c) Describe the two authentication schemes used in entity authentication in brief. CO5 (06)
